



Dr. B. C. Roy Engineering College, Durgapur

A MINI PROJECT REPORT ON

BI-DIRECTIONAL PEOPLE COUNTING SYSTEM

IOT-BASED OCCUPANCY MONITORING USING ESP32 & IR SENSORS

Embedded Systems Mini-Project (4th Sem) of

BACHELOR OF TECHNOLOGY IN ELECTRICAL ENGINEERING

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CERTIFICATE:

This is to certify that the mini project report entitled “**BI-DIRECTIONAL PEOPLE COUNTING SYSTEM**” is a bonafide work carried out by:

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of the **Department of Electrical Engineering, Dr. B.C. Roy Engineering College**, during the academic session **2025–2026**, under my guidance and supervision.

The project has been completed as a part of the requirements for the **Embedded Systems Mini Project** in the **4th Semester of Bachelor of Technology in Electrical Engineering**.

The work presented in this report is based on the design and development of an **IoT-based bi-directional people counting system using ESP32 and IR sensors**, designed to monitor the number of people entering and leaving a defined space in real time.

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DECLARATION:

We hereby declare that the mini project report entitled “**BI-DIRECTIONAL PEOPLE COUNTING SYSTEM**” submitted to the **Department of Electrical Engineering, Dr. B.C. Roy Engineering College**, is our original work carried out as part of the requirements for the **4th Semester of Bachelor of Technology in Electrical Engineering** during the academic session **2025–2026**.

We further declare that this project work has been carried out by us under the guidance and supervision of **Dr. Tushnik Sarkar** and has not been submitted previously, in whole or in part, for the award of any degree, diploma, or other academic qualification.

The information, design, implementation, and results presented in this report are based on the work carried out by our project team.

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Date: 10/09/2026

Project Title:

BI-DIRECTIONAL PEOPLE COUNTING SYSTEM



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We would like to express our sincere gratitude to everyone who supported and guided us in the successful completion of our mini project entitled “**BI-DIRECTIONAL PEOPLE COUNTING SYSTEM.**”

First and foremost, we would like to express our heartfelt thanks to our project guide, **Dr. Tushnik Sarkar**, for his valuable guidance, encouragement, constructive suggestions, and continuous support throughout the development of this project. His guidance helped us to understand the practical aspects of designing and implementing the system.

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ABSTRACT:

The **Bi-Directional People Counting System** is an IoT-based system designed to automatically monitor the number of people entering and leaving a defined space in real time. The system uses an **ESP32 microcontroller** and two **IR obstacle sensors** to detect the direction of movement based on the sequence in which the sensors are triggered.

When a person crosses the sensors in the sequence **IR1** → **IR2**, the system identifies the movement as an **entry** and increases the people count by one. Conversely, when the sequence is **IR2** → **IR1**, the movement is identified as an **exit**, and the count is decreased by one, with the count prevented from going below zero. A state-based detection and waiting mechanism is used to avoid multiple counting of the same person.

The current count is displayed locally through a **16×2 LCD**, while the ESP32 also provides a **local Wi-Fi web dashboard** for remote monitoring. The dashboard displays the current number of people, the status of both IR sensors, the ESP32 IP address, and Wi-Fi signal strength.

The system provides a simple and cost-effective approach to **real-time occupancy monitoring**, reducing dependence on manual counting and providing useful information for monitoring space utilization and crowd levels. The project can be further extended with cloud connectivity and historical data analysis for more advanced monitoring applications.

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1. INTRODUCTION:

The **Bi-Directional People Counting System** is an automated system developed to accurately monitor the number of people entering and leaving a defined space in real time. The system is designed around an **ESP32 microcontroller** and **two IR obstacle sensors**, which detect the movement of a person based on the sequence in which the sensors are interrupted.

The basic principle of the system is based on **directional detection**. When a person crosses the two sensors in the sequence **IR1 → IR2**, the system recognizes the movement as an entry and increases the count. When the sequence occurs in the opposite direction, **IR2 → IR1**, the system recognizes it as an exit and decreases the count.

The ESP32 acts as the main processing unit and also provides wireless connectivity through a **local Wi-Fi web server**. The current occupancy can therefore be monitored in two ways: through a local display and through a web-based dashboard. The dashboard provides information such as the current people count, sensor status, ESP32 IP address, and Wi-Fi signal strength.

A state-based detection mechanism is incorporated to reduce false or repeated counting. After a movement is detected, the system waits until both sensors become clear before preparing to detect the next person.

The proposed system provides a simple and practical solution for **real-time occupancy monitoring and people-flow management**. It can be useful in locations where knowing the number of people inside a particular area is important for monitoring, safety, and efficient space utilization.

2. PROBLEM STATEMENT:

In many buildings and enclosed spaces, it is important to know the **number of people currently present**. Traditional manual counting methods are time-consuming and can be affected by human error. The project addresses this problem by providing an automated system capable of detecting both **entry and exit movements** and maintaining the occupancy count in real time.

2.1 Energy Wastage

Lighting and HVAC systems may continue operating at full capacity even when rooms are empty or sparsely occupied. This can result in unnecessary electrical energy consumption and increased operating costs.

2.2 Crowd Management

Manual monitoring of people entering and leaving an area is prone to human error. Without an automated counting system, it becomes difficult to monitor occupancy and maintain appropriate capacity limits effectively.

2.3 Lack of Occupancy Data

Without automatic data collection, building or facility managers have limited information about occupancy patterns and peak usage periods. This makes it difficult to optimize space utilization and resource allocation.

2.4 Proposed Solution

The proposed **Bi-Directional People Counting System** uses two IR sensors and an ESP32 microcontroller to determine the direction of movement. The sequence **IR1 → IR2** represents entry, while **IR2 → IR1** represents exit. The resulting count is displayed locally and can also be monitored through the ESP32's wireless web dashboard.

3. OBJECTIVES:

The main objective of the **Bi-Directional People Counting System** is to develop an automated and reliable system for monitoring the number of people entering and leaving a defined area in real time.

The major objectives of the project are:

3.1 Automatic People Counting

To automatically determine the number of people, present in a defined space without requiring manual counting.

3.2 Bi-Directional Movement Detection

To identify whether a person is **entering or exiting** by analyzing the sequence of the two IR sensors:

- **IR1 → IR2:** Entry
- **IR2 → IR1:** Exit

3.3 Real-Time Occupancy Monitoring

To maintain and display the current number of people inside the monitored area in real time.

3.4 Local Display

To provide immediate on-site information through an LCD display showing the current people count and system status.

3.5 IoT-Based Web Monitoring

To provide a wireless local web dashboard through the ESP32 for monitoring the people count and sensor status remotely over the local Wi-Fi network.

3.6 Reduction of False Counting

To use timing, sensor sequencing, debouncing, and a state-based detection mechanism to reduce repeated or incorrect counts.

3.7 Future Expandability

To develop a system that can be further extended with cloud integration and predictive analytics for historical occupancy monitoring and peak-usage prediction.

4. METHODOLOGY:

The methodology of the **Bi-Directional People Counting System** involves designing, developing, integrating and testing an ESP32-based system capable of detecting the direction of movement and maintaining the number of people inside a defined area. The system uses two IR obstacle sensors positioned at different sensing points and an ESP32 as the main processing and communication unit.

4.1 System Planning

The first step is to define the sensing arrangement and the method of determining the direction of movement. Two IR sensors are placed along the entrance/exit path so that a person activates them sequentially.

4.2 Sensor Interfacing

The two IR sensors are connected to the ESP32. In the implemented system:

- **IR Sensor 1** → **GPIO 18**
- **IR Sensor 2** → **GPIO 19**

The sensors provide input signals to the ESP32 whenever movement is detected.

4.3 Direction Detection

The ESP32 determines the direction by checking which sensor is activated first:

IR1 → IR2 → ENTRY → Count + 1

IR2 → IR1 → EXIT → Count – 1

This sequence-based method forms the main counting principle of the project.

4.4 State-Based Processing

The software uses different states to process the sensor sequence:

- **STATE_IDLE** – Waits for a new sensor event.
- **STATE_IR1_FIRST** – IR1 is detected first and the system waits for IR2.
- **STATE_IR2_FIRST** – IR2 is detected first and the system waits for IR1.
- **STATE_WAIT_CLEAR** – Waits until both sensors become clear after a valid movement.

This method helps prevent repeated counting of the same movement.

4.5 Timeout and Debouncing

Timing and debounce mechanisms are incorporated into the software. If the required second sensor is not detected within the specified sequence time, the incomplete sequence is cancelled and the system returns to the **IDLE** state.

4.6 Count and Display Update

After a valid entry or exit is detected, the ESP32 updates the people count and displays the information through the local I²C LCD. The system also provides status information through the web interface.

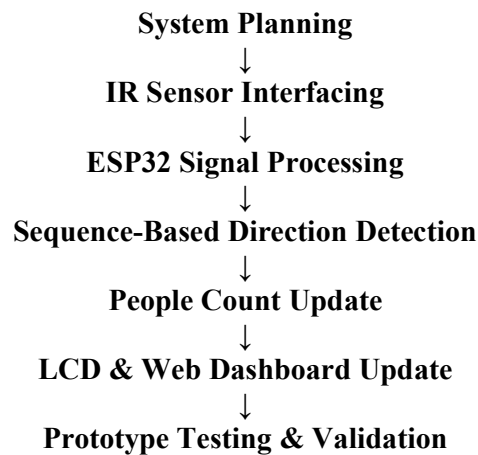
4.7 IoT Monitoring

The ESP32 hosts a local web server. The dashboard receives system data through the `/data` endpoint and displays the current people count, IR sensor status, ESP32 IP address and Wi-Fi RSSI.

4.8 Prototype Assembly and Testing

Finally, the electronic components are assembled inside a PVC enclosure. The complete system is powered, the sensors are tested for sequential detection, and the entry/exit counting operation is verified through the display and web dashboard.

Methodology Flow



This methodology provides a systematic approach for developing the proposed **ESP32-based bi-directional people counting and occupancy monitoring system**.

5. LITERATURE REVIEW:

People counting and occupancy monitoring are important applications in smart buildings and automated monitoring systems. Conventional methods generally depend on manual observation or simple counting techniques, which may not provide reliable real-time information. The proposed project focuses on an automated approach using **IR sensors and an ESP32 microcontroller**.

The project materials describe the use of two IR sensors to determine the direction of movement. The sequence **IR1** → **IR2** is interpreted as entry, while **IR2** → **IR1** is interpreted as exit. This directional approach allows the system to maintain the number of people currently present in a monitored area.

The developed system also incorporates a state-based mechanism to avoid repeated counting. After a person is detected, the system enters a **WAIT_CLEAR** state and waits until both sensors become clear before accepting the next movement. This helps prevent multiple counts caused by a single person crossing the sensors.

For real-time monitoring, the project uses the wireless capability of the ESP32. The controller hosts a **local web server**, while the web dashboard displays the current people count, IR sensor status, ESP32 IP address, and Wi-Fi signal strength.

5.1 Summary

From the available project materials, the selected approach combines:

- **Two IR obstacle sensors** for movement detection.
- **ESP32 DevKit V4** for processing and wireless communication.
- **LCD display** for local monitoring.
- **Local web dashboard** for wireless monitoring.
- **State-based detection and timing** for directional counting and reduction of false counts.

This combination provides the foundation for the proposed **Bi-Directional People Counting System** and its real-time occupancy monitoring capability.

6. HARDWARE COMPONENTS USED:

The **Bi-Directional People Counting System** is built using a microcontroller, infrared sensors, display unit, power supply, and supporting components. Each component performs a specific function in the operation of the system.

Sl. No.	Component	Quantity	Function
1	ESP32 DevKit V4	1	Main processing unit and Wi-Fi-enabled controller
2	IR Obstacle Module	2	Detects the movement of people at the entrance/exit
3	Display Unit	1	Displays the real-time people count
4	5V, 1.5A AC-DC Converter	1	Provides regulated 5V DC supply
5	5V DC Female Socket	1	Provides connection for the DC power supply
6	5V Buzzer	1	Indicates the limited Value
7	Small Switch	2	Used for ON/OFF control
8	PVC Enclosure Box	1	Provides mechanical protection and houses the circuit
9	Jumper & Hookup Wires	1 set	Used for electrical connections

The component document lists the ESP32 DevKit V4, two IR obstacle modules, display, power components, capacitors, switches, PVC enclosure, and wiring as part of the project hardware.

6.1 ESP32 DevKit V4

The **ESP32 DevKit V4** acts as the main controller of the system. It receives signals from the two IR sensors, processes their sequence and timing, maintains the people count, controls the display, and provides Wi-Fi connectivity for the local web dashboard. The project code assigns **GPIO 18 to IR Sensor 1, GPIO 19 to IR Sensor 2**, and uses **GPIO 21 and 22 for I²C communication**.

6.2 IR Obstacle Sensors

Two IR obstacle modules are used to detect a person's movement. Their activation sequence is used to determine the direction of movement. **IR1 followed by IR2 indicates entry**, whereas **IR2 followed by IR1 indicates exit**.

6.3 Display Unit

The display provides immediate local information about the current people count and system status. The project code includes functions for updating the display with the people count and messages such as **“ENTRY DETECTED”** and **“EXIT DETECTED.”**

6.4 Enclosure and Supporting Components

The power components, switches, capacitors, wires, and PVC enclosure support the safe and practical construction of the prototype. The enclosure provides physical protection and allows the complete system to be assembled as a compact unit.



7. SOFTWARE TECHNOLOGIES USED:

The **Bi-Directional People Counting System** uses embedded programming and web technologies to process IR sensor signals, maintain the people count, display the count locally, and provide real-time monitoring through a Wi-Fi web dashboard. The main software technologies used in the project are given below.

7.1 C/C++ Programming

The core control logic of the system runs on the **ESP32 microcontroller**. The program is developed using **C/C++ with the Arduino Core/ESP-IDF environment**. It controls the IR sensors, counting logic, display and web-server functions.

7.2 Hardware Interrupt Service Routines (ISRs)

Hardware interrupts are used to capture changes from the two IR sensors quickly. This allows the ESP32 to record sensor events with high timing accuracy, which is important for determining the direction of movement.

7.3 FreeRTOS

The project design uses **FreeRTOS** for task management and timing-related operations such as debouncing delays. This helps reduce false sensor detections and improves the reliability of the counting process.

7.4 Wi-Fi and Web Server

The ESP32 provides a **local Wi-Fi web server** for remote monitoring. The program uses the ESP32 Wi-Fi and web-server libraries, with the web server operating on **port 80**.

7.5 HTML5 and CSS3

HTML5 is used to structure the web dashboard, while **CSS3** is used to design its interface and presentation. The dashboard provides information such as the current people count and sensor status.

7.6 JavaScript Fetch API

JavaScript is used with the **Fetch API** to obtain updated data from the ESP32. The dashboard automatically requests new data approximately every **300 ms**, allowing real-time updates without refreshing the webpage.

7.7 JSON

JSON (JavaScript Object Notation) is used as the lightweight data format for transferring information between the ESP32 web server and the dashboard. The transmitted information includes the **people count, sensor status and ESP32 IP address**.

7.8 Libraries Used in ESP32 Program

The source program includes the following important libraries:

- **WiFi.h** – Wi-Fi connectivity
- **WebServer.h** – Local web-server functionality



- **Wire.h** – I²C communication
- **LiquidCrystal_I2C.h** – I²C LCD control

The program also defines GPIO connections for the two IR sensors and I²C communication lines.

Summary

Thus, the software architecture combines **ESP32 embedded programming, interrupt-based sensor detection, Wi-Fi communication and a browser-based HTML/CSS/JavaScript dashboard** to achieve real-time bi-directional people counting and monitoring.

8. SYSTEM ARCHITECTURE:

The **Bi-Directional People Counting System** is designed around an ESP32 microcontroller connected to two IR obstacle sensors and a local display. The ESP32 processes the sensor signals, determines the direction of movement, updates the people count, and provides the information through a local Wi-Fi web dashboard.

8.1 Main Blocks of the System

The major blocks of the system are:

1. **IR Sensor 1 (IR1)** – Detects a person crossing the first sensing point.
2. **IR Sensor 2 (IR2)** – Detects a person crossing the second sensing point.
3. **ESP32 DevKit V4** – Acts as the main processing and control unit.
4. **Display Unit** – Provides local real-time information about the people count.
5. **Wi-Fi Web Server** – Hosts the monitoring dashboard.
6. **Web Dashboard** – Displays count, sensor status, ESP32 IP address and Wi-Fi information.

8.2 Block Diagram

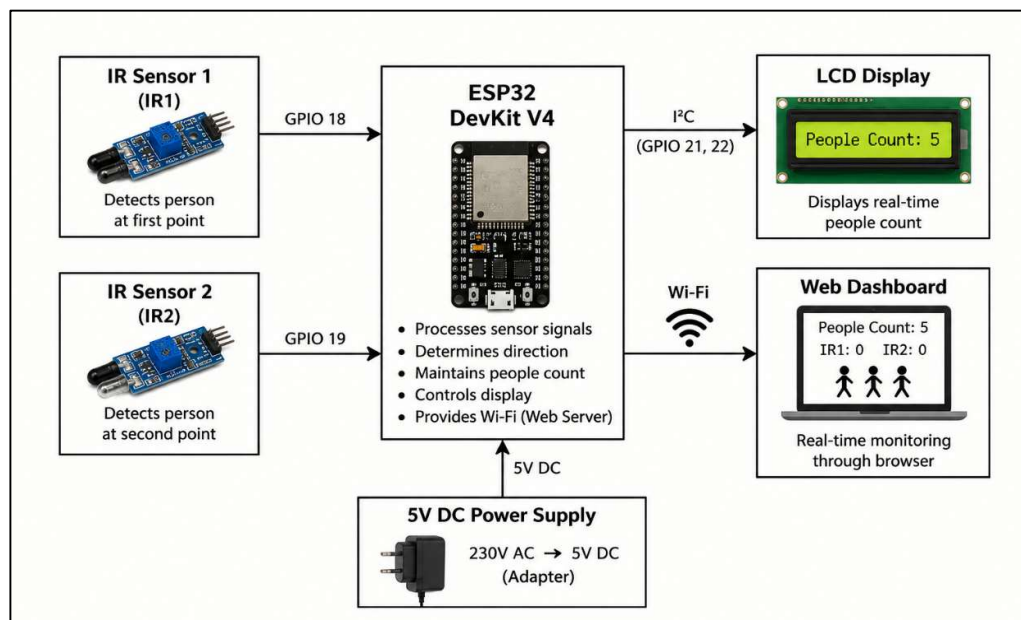


Fig 7.1: Block Diagram

8.3 Working of the Architecture

The two IR sensors are positioned at the entrance/exit so that a person activates them in a particular sequence. The ESP32 detects and processes these sensor events. The program uses separate states for **IDLE**, **IR1-FIRST**, **IR2-FIRST** and **WAIT-CLEAR**, allowing it to determine the direction of movement and avoid repeated or false counting.

The resulting count is displayed locally and is also made available through the ESP32's local web server. The dashboard provides the current count along with the status of both IR sensors and network information.



9. WORKING PRINCIPLE:

The **Bi-Directional People Counting System** works by detecting the order in which a person interrupts two infrared (IR) sensors placed along the entrance/exit path. The ESP32 processes the sensor sequence and determines whether the person is **entering or exiting** the monitored area.

9.1 Sensor Arrangement

Two IR obstacle sensors are installed at two different positions:

- **IR Sensor 1 (IR1):** First sensing point
- **IR Sensor 2 (IR2):** Second sensing point

When a person passes through the entrance, the two sensors are activated one after another. The **sequence of activation** determines the direction of movement.

9.2 Entry Detection

When a person moves **from IR1 towards IR2**, the sequence is:

IR1 → IR2

The ESP32 recognizes this sequence as an **ENTRY event** and increases the people count by **1**.

Example:

Previous Count = 5
Person enters → $5 + 1 = 6$

9.3 Exit Detection

When a person moves **from IR2 towards IR1**, the sequence is:

IR2 → IR1

The ESP32 recognizes this sequence as an **EXIT event** and decreases the people count by **1**. The count is prevented from becoming negative.

Example:

Previous Count = 6
Person exits → $6 - 1 = 5$

9.4 Waiting and Reset

After detecting an entry or exit, the system enters the **WAIT_CLEAR** state. It waits until both IR sensors become clear before accepting another movement. This prevents a single person from being counted multiple times.

9.5 Sequence Timeout

The system also uses a timeout for incomplete sensor sequences. If the second sensor is not triggered within the specified time, the sequence is cancelled and the system returns to the **IDLE** state.

Working Sequence

Person Movement → IR Sensor Detection → ESP32 Processing → Direction Identification → Count Update → Display/Web Dashboard Update

Thus, the system provides **real-time and bi-directional occupancy counting** using only the order of two IR sensor activations.



10. BI-DIRECTIONAL DETECTION LOGIC:

The main feature of the **Bi-Directional People Counting System** is its ability to identify whether a person is **entering or exiting** by analyzing the sequence in which the two IR sensors are triggered.

10.1 Entry Detection

When the person crosses **IR1 first** and then **IR2**, the system identifies the movement as an entry.

Sequence:

IR1 → IR2 → ENTRY → Count = Count + 1

The ESP32 records the first sensor event and waits for IR2 to be triggered within the permitted sequence time. If the correct sequence is detected, the entry counter is incremented.

10.2 Exit Detection

When the person crosses **IR2 first** and then **IR1**, the system identifies the movement as an exit.

Sequence:

IR2 → IR1 → EXIT → Count = Count - 1

The count is decreased only when the valid reverse sequence is detected, and the program prevents the count from going below zero.

10.3 State-Based Detection

The software uses different states to reliably process sensor sequences:

- **STATE_IDLE** – System waits for a new sensor event.
- **STATE_IR1_FIRST** – IR1 has been detected first; system waits for IR2.
- **STATE_IR2_FIRST** – IR2 has been detected first; system waits for IR1.
- **STATE_WAIT_CLEAR** – A valid movement has been detected; system waits until the sensors are clear.

10.4 Detection Table

Sensor Sequence	Movement	Counter Operation
IR1 → IR2	Entry	Count + 1
IR2 → IR1	Exit	Count - 1
IR1 only	Incomplete	No change
IR2 only	Incomplete	No change
Both sensors clear	Ready	Return to IDLE

10.5 Advantages of the Logic

This sequence-based method helps the system:

- Identify both entry and exit directions.
- Avoid counting a person from a single sensor activation.
- Reject incomplete sequences using a timeout.
- Prevent repeated counting through the **WAIT_CLEAR** state.

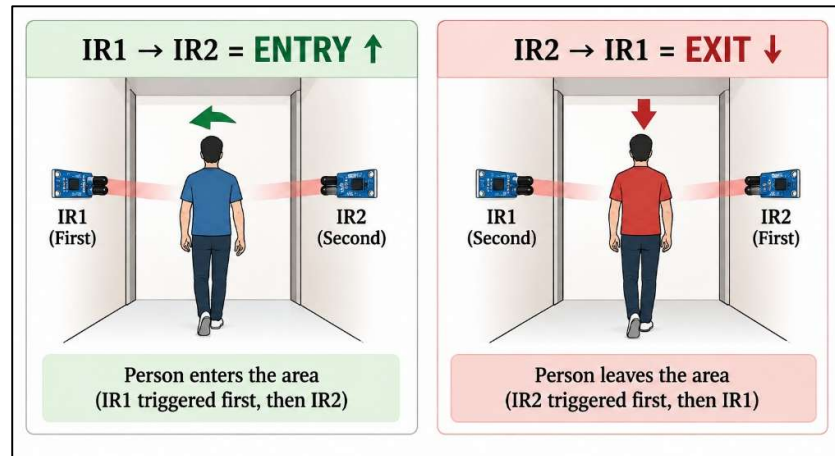


Figure 10.1: Bi-Directional Detection Sequence

11. ALGORITHM AND FLOWCHART:

The system uses a **state-based algorithm** to identify the direction of movement from the order of activation of IR1 and IR2. The program also uses a sequence timeout and a **WAIT_CLEAR** state to avoid incorrect or repeated counting.

11.1 Algorithm

Step 1: Start the ESP32 system and initialize the IR sensors, display, Wi-Fi and web server.

Step 2: Set the initial people count to zero and keep the system in the **IDLE** state.

Step 3: Continuously monitor IR1 and IR2 for sensor events.

Step 4: If **IR1 is detected first**, enter the `STATE_IR1_FIRST` state and wait for IR2.

Step 5: If IR2 is detected within the allowed sequence time, identify the movement as **ENTRY** and increase the count by 1.

Step 6: If **IR2 is detected first**, enter the `STATE_IR2_FIRST` state and wait for IR1.

Step 7: If IR1 is detected within the allowed sequence time, identify the movement as **EXIT** and decrease the count by 1.

Step 8: After a valid entry or exit, move to the **WAIT_CLEAR** state.

Step 9: Wait until both IR sensors become clear to prevent multiple counting.

Step 10: If the second sensor is not detected within the timeout period, cancel the incomplete sequence and return to **IDLE**.

Step 11: Update the local display and web dashboard with the current count.

Step 12: Repeat the process continuously.

11.2 Flowchart

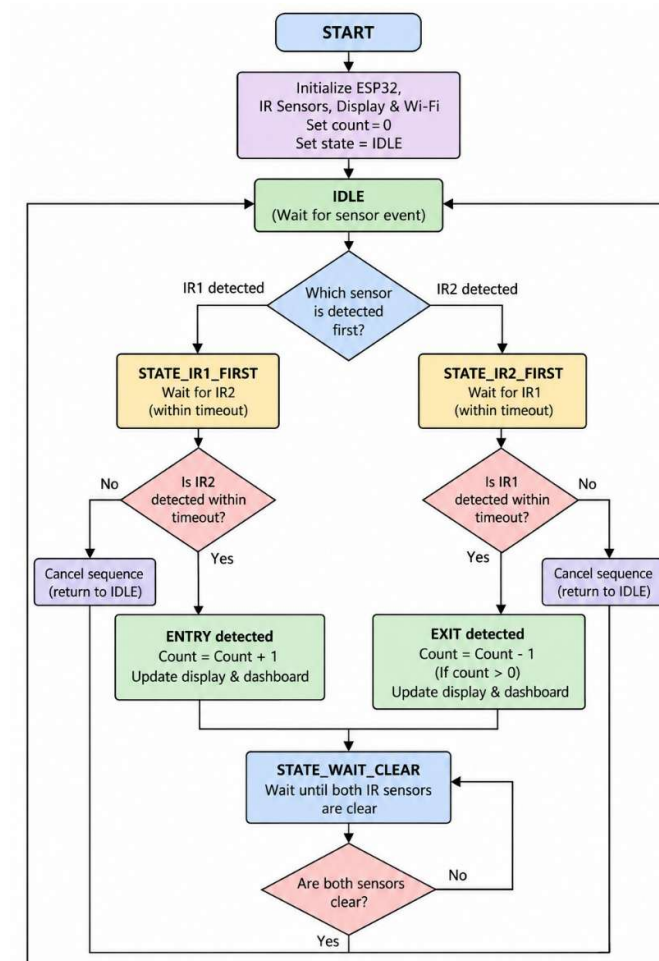


Fig. 11.1: Flowchart of the Bi-Directional People Counting Algorithm

12. HARDWARE IMPLEMENTATION:

The hardware implementation of the **Bi-Directional People Counting System** is based on an ESP32 DevKit V4, two IR obstacle sensors, a display unit, power-supply components and an enclosure. The ESP32 acts as the main processing unit and also provides the local Wi-Fi web server.

12.1 ESP32 DevKit V4

The **ESP32 DevKit V4** is the central controller of the project. It receives signals from both IR sensors, processes their activation sequence, calculates the number of people and controls the display and web interface.

The program assigns:

- **IR1 → GPIO 18**
- **IR2 → GPIO 19**
- **I²C SDA → GPIO 21**
- **I²C SCL → GPIO 22**

12.2 IR Obstacle Sensors

Two IR obstacle modules are used for detecting the movement of people at two sensing points. The first sensor detects the person at the first point and the second sensor detects the person at the second point.

The order of activation is used to identify:

- **IR1 → IR2 : Entry**
- **IR2 → IR1 : Exit**

12.3 Display Unit

A display is provided for local indication of the current people count. The project code uses an **I²C 16×2 LCD**, controlled through the `LiquidCrystal_I2C` library.

12.4 Power Supply

The hardware requires a suitable DC power source for the ESP32 and connected modules. The component list includes a **5V, 1.5A AC-DC converter**, a 5V DC female socket and supporting capacitors.

12.5 Enclosure and Wiring

The components are interconnected using jumper/hookup wires and housed inside a PVC enclosure. The enclosure provides physical protection and makes the prototype suitable for practical installation.

12.6 Hardware Connection Summary

Component	Connection / Function
IR Sensor 1	ESP32 GPIO 18
IR Sensor 2	ESP32 GPIO 19
Display I ² C SDA	GPIO 21
Display I ² C SCL	GPIO 22
ESP32	Main processing and Wi-Fi
Power Supply	Provides DC power
PVC Enclosure	Protects the assembled circuit

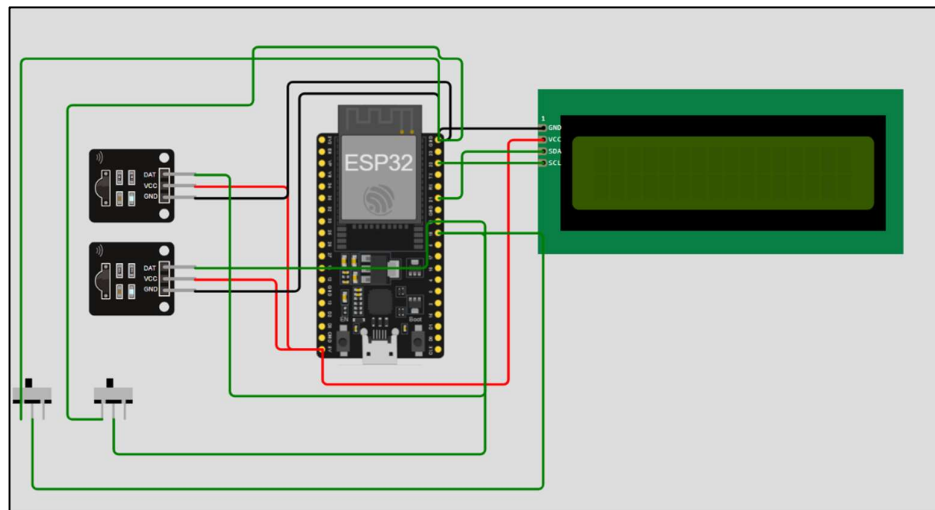


Figure 12.1: Hardware Implementation of the Bi-Directional People Counting System

13. SOFTWARE IMPLEMENTATION:

The software implementation of the **Bi-Directional People Counting System** is developed for the ESP32 using C/C++ programming. The program integrates sensor detection, direction identification, people counting, LCD display control and a local Wi-Fi web server.

13.1 Program Initialization

During startup, the ESP32 initializes the IR sensor pins, I²C communication and LCD. Hardware interrupts are attached to both IR sensors so that sensor changes can be detected immediately. The Wi-Fi connection and local web server are then started.

13.2 Sensor Event Detection

The program continuously monitors the two IR sensors. When IR1 is detected first, the system enters the `STATE_IR1_FIRST` state. Similarly, when IR2 is detected first, it enters the `STATE_IR2_FIRST` state.

13.3 Counting Operation

The software compares the timing of the second sensor event with the first sensor event.

- **IR1 → IR2:** `countEntry()` is executed and the people count increases.
- **IR2 → IR1:** `countExit()` is executed and the people count decreases.

After a valid detection, the system changes to `STATE_WAIT_CLEAR`.

13.4 Timeout and Error Handling

If the second sensor is not triggered within the specified sequence time, the incomplete sequence is cancelled. The state is reset to **IDLE**, preventing an incorrect count.

13.5 LCD Display

The program initializes the I²C LCD and displays system information such as the current count and sensor-detection messages. The LCD is initialized during the ESP32 startup sequence.

13.6 Web Server Implementation

The ESP32 hosts a local web server with two main routes:

- `/` – Displays the web dashboard.
- `/data` – Provides the current system data.

The server is started after the Wi-Fi connection is established.

13.7 Dashboard Data Updating

The JavaScript dashboard requests data from `/data` and updates the displayed **people count, ESP32 IP address, Wi-Fi RSSI and IR1/IR2 sensor status** without reloading the webpage.

Software Flow

Sensor Detection → Interrupt → State Processing → Direction Detection → Count Update → LCD Update → Web Dashboard Update

This software structure enables the ESP32 to perform real-time people counting while simultaneously providing local display and Wi-Fi-based monitoring.



14. WEB DASHBOARD AND IoT MONITORING:

The project provides a **local Wi-Fi web dashboard** through the ESP32. It allows the current occupancy information and sensor conditions to be monitored through a web browser without requiring a separate cloud server.

14.1 Local Web Dashboard

The ESP32 works as a local web server and provides the dashboard through its Wi-Fi connection. The dashboard is designed to show important system information in real time.

14.2 Information Displayed

The dashboard provides the following information:

- **Current People Count**
- **IR Sensor 1 Status**
- **IR Sensor 2 Status**
- **ESP32 IP Address**
- **Wi-Fi RSSI (signal strength)**

The sensor status is displayed as **DETECTED** or **CLEAR** depending on the current sensor condition.

14.3 Real-Time Data Communication

The dashboard uses JavaScript's **Fetch API** to request data from the ESP32 `/data` endpoint. The received JSON data is then used to update the dashboard without refreshing the webpage.

The project presentation specifies that the dashboard requests updated information approximately every **300 ms**, providing near real-time monitoring.

14.4 Data Flow

The monitoring process can be represented as:

IR Sensors → ESP32 → People Count Processing → JSON Data → Wi-Fi → Web Browser

The ESP32 therefore acts as both the **processing unit** and the **local IoT communication gateway**.

14.5 Benefits of the Dashboard

The web dashboard provides:

1. Remote viewing within the local Wi-Fi network.
2. Real-time occupancy information.
3. Direct monitoring of both IR sensors.
4. Network information for accessing the ESP32.
5. A simple browser-based interface without requiring additional monitoring hardware.



Figure 14.1: ESP32 Local Web Dashboard

15. PROTOTYPE DEVELOPMENT

The prototype of the **Bi-Directional People Counting System** is designed as a compact unit that can be installed near an entrance or exit. The arrangement places the two IR sensors at separate sensing points so that the direction of a person's movement can be determined.

15.1 Physical Arrangement

The main hardware components are assembled inside a **PVC enclosure**. The enclosure provides mechanical protection and makes the system suitable for practical installation.

The prototype consists mainly of:

- ESP32 DevKit V4
- Two IR obstacle sensor modules
- Display unit
- Power-supply section
- Switches
- Capacitors and connecting wires
- PVC enclosure

15.2 Sensor Placement

The two IR sensors are positioned at different points along the passage. When a person crosses the sensing region, the sensors are triggered sequentially.

For Entry:

IR1 → IR2 → Count increases

For Exit:

IR2 → IR1 → Count decreases

15.3 Display Arrangement

The display provides immediate local information about the system. The project code initializes an I²C LCD and uses it for system messages and people-count information.

15.4 Enclosure and Power

The electronics are housed in a PVC enclosure, while the power section supplies the required DC voltage to the controller and associated modules. The component list includes a 5V AC-DC converter, DC socket, switches, capacitors and enclosure.

15.5 Prototype Operation

After powering the system, the ESP32 initializes the sensors, display and Wi-Fi connection. It then continuously monitors the two IR sensors. When a valid sensor sequence is detected, the people count is updated and the information is made available on both the local display and web dashboard.



Figure 15.1: Prototype of the Bi-Directional People Counting System

16. RESULTS AND DISCUSSION:

The developed **Bi-Directional People Counting System** is designed to detect the movement of people through two IR sensors and determine the direction of movement from their activation sequence.

16.1 Entry Detection Result

When **IR1 is triggered first and IR2 is triggered next**, the system identifies the movement as an entry and increases the people count by one.

IR1 → IR2 → ENTRY → Count + 1

16.2 Exit Detection Result

When **IR2 is triggered first and IR1 is triggered next**, the system identifies the movement as an exit and decreases the people count by one. The program prevents the count from becoming negative.

IR2 → IR1 → EXIT → Count – 1

16.3 Multiple-Counting Prevention

After a valid entry or exit is detected, the program enters the **WAIT_CLEAR** state. It waits until the sensors become clear before processing another movement. This helps prevent a single person from being counted more than once.

16.4 Display and Monitoring Result

The current count and sensor conditions can be monitored through the local display and the ESP32 web dashboard. The dashboard provides the people count, IR1 and IR2 status, ESP32 IP address and Wi-Fi RSSI.

16.5 Overall Result

The implemented system demonstrates the following functions:

Function	Result
Entry detection	IR1 → IR2
Exit detection	IR2 → IR1
People count update	Successful sequence-based counting
False/repeated counting control	WAIT_CLEAR state
Incomplete sequence handling	Timeout and return to IDLE
Local monitoring	Display unit
Wi-Fi monitoring	ESP32 Web Dashboard

Therefore, the prototype demonstrates a practical approach for **real-time bi-directional people counting and occupancy monitoring** using ESP32 and two IR sensors.

17. ADVANTAGES:

The **Bi-Directional People Counting System** provides a simple and effective method for monitoring the number of people entering and leaving a room. The system uses two IR sensors with sequence-based detection and an ESP32 controller to determine the direction of movement. The counted value is displayed locally and can also be monitored through a Wi-Fi-based web dashboard.

17.1 Bi-Directional Counting

The system can distinguish between **entry and exit** by observing the order in which the two IR sensors are triggered:

- **IR1 → IR2:** Person entering
- **IR2 → IR1:** Person exiting

This allows the system to maintain an occupancy count rather than simply detecting movement.

17.2 Real-Time Monitoring

The ESP32 continuously processes the sensor signals and updates the people count. The current count and sensor status are also made available through the web dashboard.

17.3 Local Display

A display unit is incorporated into the system to provide a direct indication of the current people count without requiring a separate computer or mobile device. The project software uses an I2C LCD interface for this purpose.

17.4 Wireless Web Dashboard

The ESP32 hosts a local web dashboard that can display information such as:

- Current people count
- IR1 sensor status
- IR2 sensor status
- ESP32 network information
- Wi-Fi signal information

The dashboard periodically requests updated data from the ESP32 using the `/data` interface.

17.5 Reduced False Counting

The software incorporates **debouncing and timing controls** to handle rapid or unwanted sensor transitions. The detection logic also uses a **WAIT_CLEAR** state, requiring the sensors to return to the clear condition before another person is counted. This helps prevent repeated counting from a single movement.

17.6 Compact Prototype

The complete system is designed as a compact prototype with the sensing elements and display integrated into an enclosure. This makes the unit suitable for practical demonstration and installation at a doorway or passage.

17.7 Simple Hardware Structure

The system is built around an ESP32 controller, two IR obstacle sensors, a display unit, power components, switches, wiring and an enclosure. The component list provides an approximate total project cost of ₹1470, making the prototype relatively economical to construct.

17.8 Easy Expansion

Since the ESP32 provides Wi-Fi connectivity and a web interface, the present system can be extended in future with cloud-based monitoring, data logging and analytics without fundamentally changing the basic people-detection principle.

In summary, the major advantage of the proposed system is that it combines **bi-directional detection, real-time counting, local indication and wireless monitoring** in a compact ESP32-based prototype.

18. LIMITATIONS:

Although the **Bi-Directional People Counting System** provides real-time occupancy monitoring, the prototype has certain limitations that should be considered during practical implementation.

18.1 Dependence on Sensor Alignment

The accuracy of counting depends on the correct positioning and alignment of the two IR sensors. Improper placement may affect the detection sequence.

18.2 Single-Passage Detection

The system is designed around two IR sensing points and a defined sequence of beam interruptions. Therefore, simultaneous movement of multiple people through the sensing area may affect the counting accuracy.

18.3 IR Sensor Limitations

IR obstacle sensors can be affected by the physical conditions of the sensing environment. Changes in object position, distance and detection conditions can influence the sensor output.

18.4 Limited Prototype Testing

The developed system is primarily a prototype for demonstrating bi-directional people counting. Large-scale deployment would require additional testing under different environmental and operational conditions.

18.5 Local Network Dependency

The web dashboard is hosted by the ESP32 and is intended for local wireless monitoring. Therefore, access to the dashboard depends on the availability and stability of the configured Wi-Fi network.

18.6 No Long-Term Data Storage

The present implementation provides real-time information through the display and web dashboard, but it does not include a dedicated cloud database or long-term occupancy-data storage system. The project presentation identifies cloud platforms such as AWS IoT and Blynk as possible future extensions.

18.7 Counting Logic Constraints

The software uses timing, debounce and sequence conditions to identify valid entry and exit movements. Very unusual or incomplete sensor sequences may therefore not be interpreted as a valid entry or exit.

18.8 Future Improvement Required

For larger installations, the system could be improved through better sensing arrangements, cloud connectivity, data logging and predictive analytics. These improvements would make the system more suitable for continuous occupancy monitoring and analysis.

Overall, the limitations mainly arise from the prototype's two-sensor arrangement, environmental sensing conditions, local-network operation and the absence of long-term cloud-based data storage.



19. APPLICATIONS:

The **Bi-Directional People Counting System** can be used in places where monitoring the number of people entering and leaving a particular area is required. By combining IR-based movement detection with ESP32 processing and wireless monitoring, the system can provide useful real-time occupancy information.

19.1 Classrooms and Laboratories

The system can be installed at the entrance of classrooms and laboratories to monitor the number of people present inside the room.

19.2 Offices

It can be used in office spaces to keep track of occupancy and provide an indication of the number of people currently inside an area.

19.3 Libraries

The system can help monitor the number of people entering and leaving a library, which can be useful for basic occupancy management.

19.4 Seminar Halls and Meeting Rooms

It can be applied at the entrance of seminar halls and meeting rooms to monitor occupancy during events and sessions.

19.5 Shops and Small Commercial Spaces

The system can provide a simple method of tracking the number of visitors entering and leaving small commercial areas.

19.6 Public Buildings

The prototype can be adapted for occupancy monitoring in public buildings where information about the number of people inside a particular section is useful.

19.7 Energy Management

Occupancy information can be used as an input for future **automatic energy-management systems**. For example, lighting or ventilation systems could be controlled according to the number of people present.

19.8 IoT-Based Monitoring

Since the ESP32 provides Wi-Fi connectivity and a local web dashboard, the system can form the basis of a larger IoT monitoring solution. The project presentation also identifies **AWS IoT and Blynk** as possible future extensions.

19.9 Smart Building Systems



The people-counting information can be integrated into future smart-building systems for occupancy monitoring, resource management and automated control.

Overall, the proposed system is suitable for applications where **entry, exit and occupancy information** needs to be monitored using a compact and relatively simple electronic system.

20. FUTURE SCOPE:

The present **Bi-Directional People Counting System** provides a basic platform for real-time occupancy monitoring using ESP32 and IR sensors. The system can be further improved by adding advanced IoT and data-analysis features.

20.1 Cloud Connectivity

The system can be connected to cloud platforms such as **AWS IoT or Blynk** for remote monitoring and data collection.

20.2 Data Logging

Future versions can store occupancy data over longer periods. This would allow users to study daily, weekly and monthly patterns of room usage.

20.3 Predictive Analytics

Historical occupancy data can be analyzed to predict future occupancy levels and identify periods of high or low usage. The project presentation specifically identifies **predictive analytics and forecasting** as future development areas.

20.4 Smart Energy Management

The occupancy information can be integrated with building automation systems so that lighting, ventilation or other electrical loads can be controlled according to the number of people present.

20.5 Improved Sensing

The sensing arrangement can be further developed to improve reliability when multiple people pass through the entrance or when movement occurs at unusual speeds or positions.

20.6 Mobile Monitoring

A future version can provide a dedicated mobile interface or application so that occupancy information can be viewed conveniently from smartphones.

20.7 Safety and Occupancy Alerts

The system can be extended to generate alerts when the number of people in a particular area reaches a predefined limit.

Overall, the project can be developed from a basic local people-counting prototype into a more advanced **IoT-based smart occupancy monitoring system** with cloud connectivity, data logging, forecasting and automated building control.

21. COST ESTIMATION:

The approximate cost of the components used for developing the **Bi-Directional People Counting System** is given below. The costing is based on the component list provided for the project.

Sl. No.	Component	Quantity	Approx. Cost (₹)
1	ESP32 DevKit V4	1	447.22
2	IR Obstacle Module	2	60
3	OLED Display	1	209
4	5V, 1.5A AC-DC Converter	1	120
5	5V DC Female Socket	1	20
6	Capacitor (4.7 μ F)	4	40
7	Small Switch	2	20
8	PVC Enclosure Box	1	200
9	Jumper & Hook-up Wires	1 set	50
10	Additional/Other Cost	—	200
	Total Approximate Cost		₹1470

The component document gives an **approximate total project cost of ₹1470**, with an approximate **per-head cost of ₹245** for the six-member team.

21.1 Cost Analysis

The **ESP32 DevKit V4** represents the major component cost because it performs sensor processing, counting logic and Wi-Fi-based web-server functions. The remaining cost is distributed among the sensing modules, display, power components, enclosure and wiring.

The project therefore demonstrates that a functional prototype for bi-directional people counting can be developed using a relatively small set of electronic components and an ESP32-based control system.

Note: The above table follows the supplied component-cost document. The project PPT/code describe the display and power arrangement somewhat differently, so the final submitted report should retain the BOM as supplied unless the hardware list is officially revised.

22. CONCLUSION:

The **Bi-Directional People Counting System** is an ESP32-based prototype developed to monitor the number of people entering and leaving a defined area. The system uses two IR sensors to identify the direction of movement by analyzing the sequence in which the sensors are triggered. An **IR1 → IR2** sequence is treated as an entry, while an **IR2 → IR1** sequence is treated as an exit.

The ESP32 acts as the main controller and processes the sensor signals using programmed detection logic. The software incorporates timing and debounce mechanisms along with a **WAIT_CLEAR** state to ensure that a completed movement is properly processed before another movement is counted.

The system also provides local indication and a Wi-Fi-based web interface for monitoring the current count and sensor status. This makes the prototype useful for demonstrating real-time occupancy monitoring using embedded and IoT technologies.

The project demonstrates the practical application of **ESP32, IR sensing, embedded programming and wireless web monitoring** in a single system. With further development, the system can be extended with cloud connectivity, data logging, predictive analytics and smart energy-management features.

Final Statement

Thus, the developed system successfully establishes a foundation for **automated, bi-directional people counting and real-time occupancy monitoring**, while providing scope for future development into a more advanced IoT-based smart monitoring solution.

23. REFERENCES:

The following references were used for the development and documentation of the **Bi-Directional People Counting System**:

1. **ESP32 DevKit V4 Documentation** – Technical information related to ESP32-based embedded development and Wi-Fi connectivity.
2. **Arduino ESP32 Documentation** – Reference material for programming ESP32 using the Arduino framework.
3. **ESP32 Source Code developed for the project** – Used for understanding the sensor interfacing, counting algorithm, interrupt handling, display operation and web dashboard implementation.
4. **Project Presentation: “Bi-Directional People Counting System – IoT-Based Occupancy Monitoring using ESP32 & IR Sensors”** – Used as the primary reference for the project concept, objectives, architecture, working principle and future scope.
5. **Project Component and Cost List** – Used for preparing the hardware component and cost-estimation section of the report.
6. **IR Obstacle Sensor Module Reference** – Used for understanding the basic operation of IR-based object detection and sensor interfacing.
7. **I²C Display Interface Reference** – Used for understanding communication between the ESP32 and the display module.
8. **HTML, CSS and JavaScript References** – Used for understanding the implementation of the local web dashboard and periodic data updates through the ESP32 web server.

23.1 Project Documents

- Project hardware/component list
- ESP32 source-code document
- Project presentation
- Hardware prototype and implementation details

24. PROJECT SUMMARY:

The **Bi-Directional People Counting System** is an ESP32-based occupancy monitoring prototype designed to identify and count people moving through a defined passage.

The system uses **two IR sensors** to detect movement and determine its direction. The sequence of sensor activation is used to distinguish between entry and exit. The ESP32 processes these signals and maintains the current people count.

The developed software also includes timing and debounce mechanisms to improve the reliability of sensor detection. The **WAIT_CLEAR** condition ensures that the sensing area is cleared before another movement is processed.

The system provides **real-time monitoring** through a local display and a Wi-Fi-based web dashboard. The web interface can present the current count and sensor-related information obtained from the ESP32.

The project combines the following major technologies:

- ESP32 microcontroller
- IR-based object detection
- Bi-directional counting logic
- Embedded C/C++ programming
- Local display interface
- Wi-Fi communication
- HTML, CSS and JavaScript web dashboard

The project demonstrates how embedded systems and IoT technologies can be combined to create a practical **occupancy monitoring solution**. The system can be further developed by adding cloud connectivity, long-term data storage and predictive analytics.

24.1 Key Outcome

The developed prototype establishes a working concept for **automated entry/exit detection and people counting**, providing both local and wireless access to occupancy information.



25. ACKNOWLEDGEMENT OF PROJECT WORK:

The successful completion of the **Bi-Directional People Counting System** involved the application of knowledge from embedded systems, sensors, microcontrollers, programming and IoT technologies.

The project work provided practical experience in:

- Interfacing **IR sensors** with an ESP32 microcontroller.
- Developing logic for **bi-directional movement detection**.
- Implementing real-time people-counting functionality.
- Using timing and debounce techniques for sensor signal processing.
- Interfacing a display with the controller.
- Developing a **Wi-Fi-based web dashboard** for monitoring system data.
- Designing and assembling the prototype hardware inside an enclosure.

Through this project, the team gained practical understanding of how hardware and software can be integrated to develop an automated monitoring system. The project also provided experience in troubleshooting, testing and improving the overall system operation.

The work demonstrates the practical application of electrical and embedded-system concepts in developing an **IoT-based occupancy monitoring prototype**.

26. TEAM CONTRIBUTION:

The **Bi-Directional People Counting System** was developed as a team project. Each member contributed to different activities involved in planning, development, implementation, testing and documentation of the project.

26.1 Major Areas of Contribution:

Sl. No.	Team Member	Contribution
1	Chanchal Bhattacharjee	Project planning and documentation support
2	Debasish Dutta	Hardware and software support
3	Debayan Laha	Sensor interfacing and testing support, Hardware implementation, programming Support, Documentation,
4	Sanchalak Samanta	Documentation support
5	Tushar Roy	Web dashboard, system development, hardware implementation, programming, testing and troubleshooting
6	Siddhartha Dey	Testing and presentation support

26.2 Teamwork

The project required coordination between hardware and software activities. The team worked together to understand the requirements, assemble the prototype, develop the control program, test the bi-directional counting operation and prepare the project documentation.

The combined effort of all team members helped in developing the prototype and presenting the working concept of an **ESP32-based bi-directional people counting system**.

